Physical Chemistry (II)

Lecture 21

Diatomic Molecules

Last class

Born-Oppenheimer Approx.

- Nuclei are much more massive than electrons
- Electrons adjust instantaneously to any nuclear motion

- For the fixed nuclear geometry $\vec{R}_1, \vec{R}_2, \cdots$, solve the Schrödinger equation:

$$\hat{H}_{\mathrm{el}}\psi_{\mathrm{el}}\left(\vec{r}_1, \vec{r}_2, \cdots; \vec{R}_1, \vec{R}_2, \cdots\right)$$
$$= E_{\mathrm{el}}\left(\vec{R}_1, \vec{R}_2, \cdots\right)\psi_{\mathrm{el}}\left(\vec{r}_1, \vec{r}_2, \cdots; \vec{R}_1, \vec{R}_2, \cdots\right)$$

- To solve this, one can apply any electronic structure calculation method

Last class

Orbital Picture

H atom	1-electron wf (“orbitals”)	H_2^+ molecule
He atom	2 electrons in orbitals	H_2 molecule
Li, Be, …	*N* electrons in orbitals	He_2^+, He_2, …

Not necessarily precise (except H atom)

Last class

H_2^+ System (Variational)

- Energy

$$E_{\pm} = \frac{H_{\mathrm{AA}} \pm H_{\mathrm{AB}}}{1 \pm S_{\mathrm{AB}}}$$

- Wavefunction

$$\phi_{\pm} = \frac{1}{\sqrt{2(1 \pm S_{\mathrm{AB}})}}(\psi_{1s,\mathrm{A}} \pm \psi_{1s,\mathrm{B}})$$

where

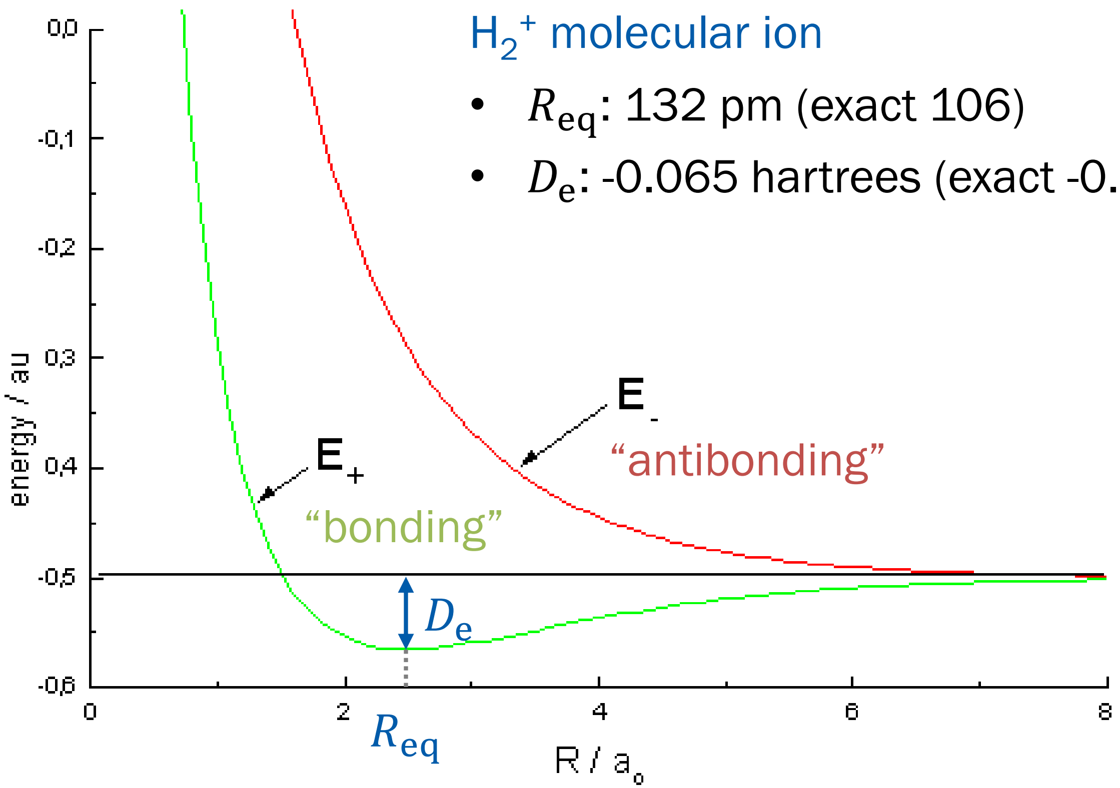
$$H_{ij} = \int \psi_{1s,i} \hat{H} \psi_{1s,j} \, \mathrm{d}\vec{r} \text{ and } S_{\mathrm{AB}} = \int \psi_{1s,\mathrm{A}} \psi_{1s,\mathrm{B}} \, \mathrm{d}\vec{r}$$

Last class

“Orbital Energies”

H_2^+ molecular ion

- R_{eq}: 132 pm (exact 106)
- D_{e}: -0.065 hartrees (exact -0.103)

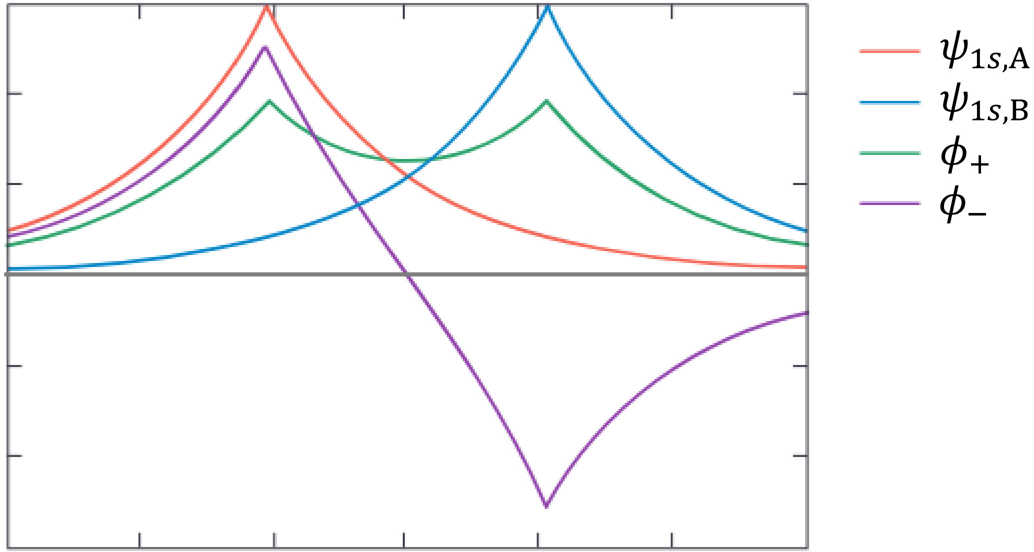


(Image: http://www.pci.tu-bs.de/aggericke/PC4e/Kap_II/H2-Ion.htm)

Last class

"Molecular Orbitals"

$$\phi_{\pm} = \frac{1}{\sqrt{2(1 \pm S_{\mathrm{AB}})}}(\psi_{1s,\mathrm{A}} \pm \psi_{1s,\mathrm{B}})$$



(Image: https://chem.libretexts.org/@go/page/1974)

Last class

Ground-State Wavefunction for H_2

- Use H_2^+ molecular orbitals

$$\phi_{\pm} = \frac{1}{\sqrt{2(1 \pm S_{\mathrm{AB}})}}(\psi_{1s,\mathrm{A}} \pm \psi_{1s,\mathrm{B}})$$

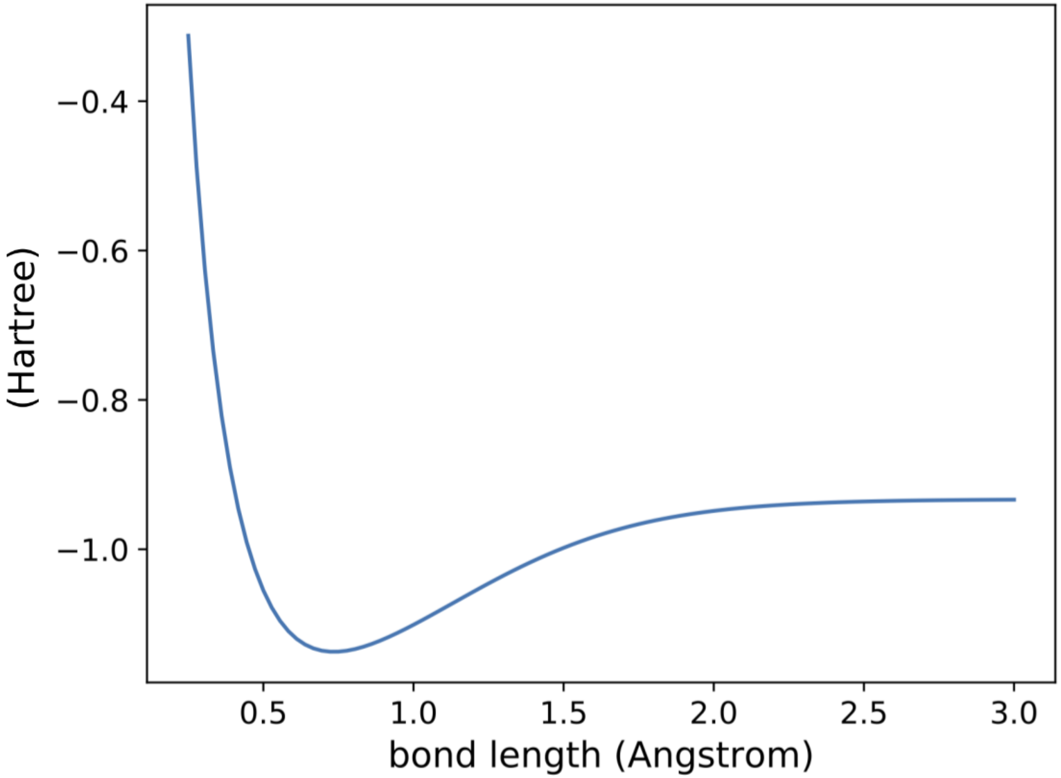
- Slater determinant (both electrons reside in ϕ_+)

$$\psi = \frac{1}{\sqrt{2}}\begin{vmatrix} \phi_+(1)|\alpha\rangle_1 & \phi_+(1)|\beta\rangle_1 \\ \phi_+(2)|\alpha\rangle_2 & \phi_+(2)|\beta\rangle_2 \end{vmatrix}$$

$$\psi = \underbrace{\phi_+(1)\phi_+(2)}_{\text{spatial (MO)}} \times \underbrace{\frac{1}{\sqrt{2}}(|\alpha\rangle_1|\alpha\rangle_2 - |\beta\rangle_1|\beta\rangle_2)}_{\text{spin}}$$

Last class

Ground-State Energy of H_2



(Image: https://quantaggle.com/competitions/gs-pes/)

Last class

Valence Bond Theory

H_2 ground state = two ground-state H atoms

$$\psi_{\mathrm{VB}} = \frac{1}{\sqrt{2(1+S^2)}}\left[\psi_{1s,\mathrm{A}}(1)\psi_{1s,\mathrm{B}}(2) + \psi_{1s,\mathrm{B}}(1)\psi_{1s,\mathrm{A}}(2)\right]$$

We can similarly obtain energy by perturbation theory

Last class

MO and VB Theories

- MO theory

$$\underbrace{\psi_{1s,\mathrm{A}}(1)\psi_{1s,\mathrm{A}}(2)}_{\text{ionic}} + \underbrace{\psi_{1s,\mathrm{A}}(1)\psi_{1s,\mathrm{B}}(2) + \psi_{1s,\mathrm{B}}(1)\psi_{1s,\mathrm{A}}(2)}_{\text{covalent}} + \underbrace{\psi_{1s,\mathrm{B}}(1)\psi_{1s,\mathrm{B}}(2)}_{\text{ionic}}$$

- VB theory

$$\psi_{1s,\mathrm{A}}(1)\psi_{1s,\mathrm{B}}(2) + \psi_{1s,\mathrm{B}}(1)\psi_{1s,\mathrm{A}}(2)$$

- Better model: somewhere between MO and VB

$$\psi = c_1\psi_{\mathrm{VB}} + c_2\psi_{\mathrm{ionic}}$$

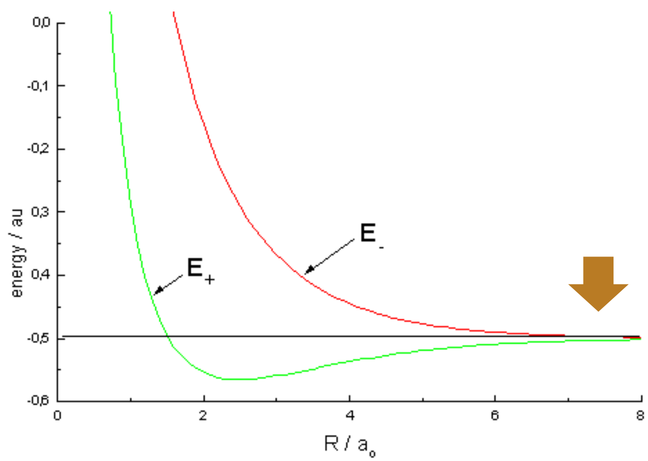
Last class

Overlap and Chemical Bond

To make a bond from two atomic orbitals,

we need a significant overlap between the two

$$S_{\mathrm{AB}} = \int \psi_{\mathrm{A}}^{*} \psi_{\mathrm{B}} \, \mathrm{d}\vec{r} \neq 0$$



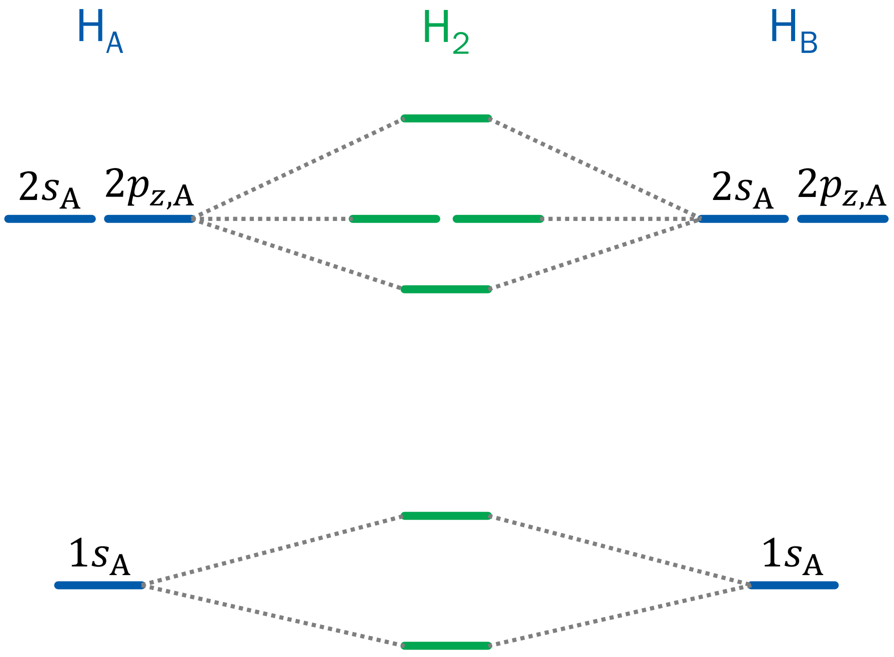
(Image: http://www.pci.tu-bs.de/aggericke/PC4e/Kap_II/H2-Ion.htm)

Last class

Molecular Orbitals

One can combine any number of atomic orbitals

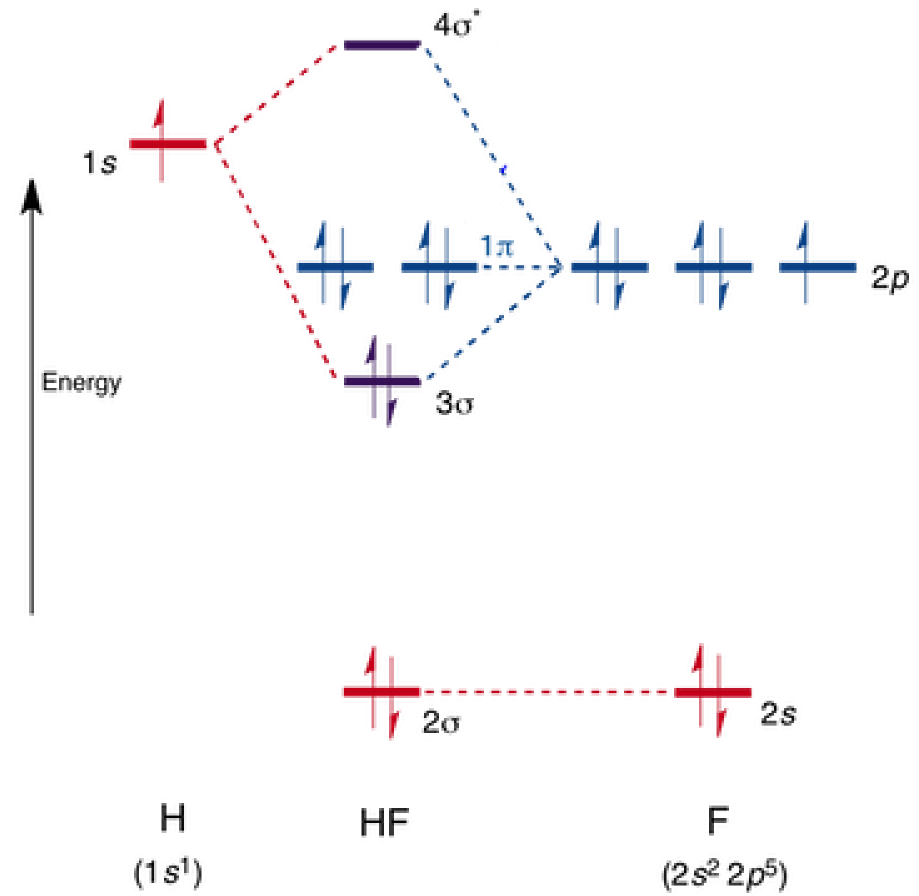
to construct molecular orbitals (LCAO-MO)



* for fixed nuclear geometry

Last class

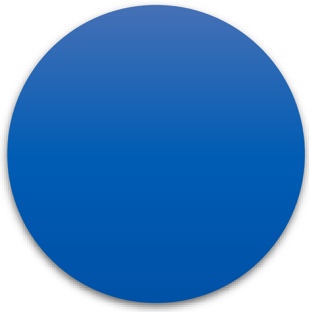
Heteronuclear Case: HF



(Image: https://www.chemtube3d.com/orbitalshf/)

Electronic States of Molecules

- Determines the energy levels
- Described by molecular term symbols



Atom

Spherical symmetry

$\vec{L}^2_{\text{total}}$ conserved

L is a good quantum number

Diatomic molecule

Cylindrical symmetry

$L_{\text{total},z}$ conserved

M_L is a good quantum number

Molecular Term Symbol

$$^{2S+1}[|M_L|]$$

- Symbols for $[|M_L|]$

$\|M_L\| =$	0	1	2	3
	Σ	Π	Δ	Φ

- $M_L = m_{l1} + m_{l2} + \cdots, M_S = m_{s1} + m_{s2} + \cdots$
- Completely filled orbitals do not contribute

Ground State of H_2

- Electron configuration: $(1\sigma_g)^2$

- $m_{l1} = 0,\ m_{l2} = 0 \ \rightarrow \ M_L = m_{l1} + m_{l2} = 0$
- $m_{s1} = +1/2\,,\ m_{s2} = -1/2 \ \rightarrow \ M_S = m_{s1} + m_{s2} = 0$
- The only possible $M_S = 0$: $S = 0$

- Hence, the term symbol is $^1\Sigma$

Ground State of H_2^- and He_2^+

- Electron configuration: $(1\sigma_g)^2(1\sigma_u)^1$

- $m_{l1} = 0,\ m_{l2} = 0,\ m_{l3} = 0 \ \rightarrow \ M_L = 0$
- $m_{s1} = +1/2\,,\ m_{s2} = -1/2\,,\ m_{s3} = \pm 1/2$

  $\rightarrow \ M_S = m_{s1} + m_{s2} + m_{s3} = \pm 1/2$
- Possible $M_S = \pm 1/2$: $S = 1/2$

- Hence, the term symbol is $^2\Sigma$

Excited State of H_2

- Electron configuration: $\left(1\sigma_g\right)^1(1\sigma_u)^1$

- $m_{l1} = 0,\ m_{l2} = 0 \ \rightarrow \ M_L = m_{l1} + m_{l2} = 0$
- $m_{s1} = \pm 1/2\,,\ m_{s2} = \pm 1/2$

  $\rightarrow \ M_S = m_{s1} + m_{s2} = \pm 1, 0$ (doubly degenerate)
- $S = 1$ and $S = 0$ can explain possible M_S values

- Hence, the possible term symbols are ${}^3\Sigma$ and ${}^1\Sigma$

Procedure: Equivalent Electrons

- Pauli exclusion principle: (a, a) impossible
- Indistinguishability: (a, b) = (b, a)

1. Construct a table for microstates

$$M_L = m_{l1} + m_{l2} + \cdots \,, \; M_S = m_{s1} + m_{s2} + \cdots$$

2. Eliminate the states that violate the two conditions

Total number of microstates = $G!/[N!\,(G-N)!]$

3. Deduce the possible values of S and $|M_L|$ (start from largest M_S and then largest M_L)

B_2 Molecule

$$(1\sigma_g)^2(1\sigma_u)^2(2\sigma_g)^2(2\sigma_u)^2(1\pi_{ux})^1(1\pi_{uy})^1$$

Possible $M_L = 0, \pm 2$, possible $M_S = 0, \pm 1$

		M_S		
		1	0	-1
M_L	2		$(1^+, 1^-)$	
	0	$(1^+, -1^+)$	$(1^+, -1^-)$; $(-1^+, 1^-)$	$(1^-, -1^-)$
	-2		$(-1^+, -1^-)$	

Total number of microstates = 4!/(2!2!) = 6

Deducing S and $|M_L|$

		M_S		
		1	0	-1
M_L	2		⊗	
	0	⊗	⊗ ⊗	⊗
	-2		⊗	

- $M_S = 0, \pm 1$ and $M_L = 0 \rightarrow S = 1$ and $|M_L| = 0$ ($^3\Sigma$)
- $M_S = 0$ and $M_L = \pm 2 \rightarrow S = 0$ and $|M_L| = 2$ ($^1\Delta$)
- $M_S = 0$ and $M_L = 0 \rightarrow S = 0$ and $|M_L| = 0$ ($^1\Sigma$)

Ground State of B_2

For the electron configuration

$$(1\sigma_g)^2(1\sigma_u)^2(2\sigma_g)^2(2\sigma_u)^2(1\pi_{ux})^1(1\pi_{uy})^1,$$

we have three possible term symbols: ${}^3\Sigma$, ${}^1\Delta$, and ${}^1\Sigma$

According to Hund's rule 1, the ground state is ${}^3\Sigma$.

Symmetry of Term Symbol

- g/u: sign change upon inversion
  - can be obtained from molecular orbitals

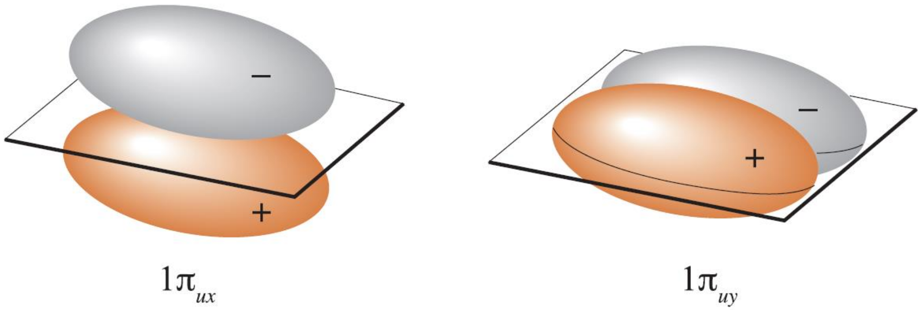
$$g \cdot g = g, \qquad g \cdot u = u, \qquad u \cdot u = g$$

- +/-: sign change upon reflection through an internuclear plane (only applies to Σ electronic states!)
  - can be obtained from molecular orbitals

$$(+)(+) = (+), \quad (+)(-) = (-), \quad (-)(-) = (+)$$

Example of B_2

$$(1\sigma_g)^2(1\sigma_u)^2(2\sigma_g)^2(2\sigma_u)^2(1\pi_{ux})^1(1\pi_{uy})^1$$

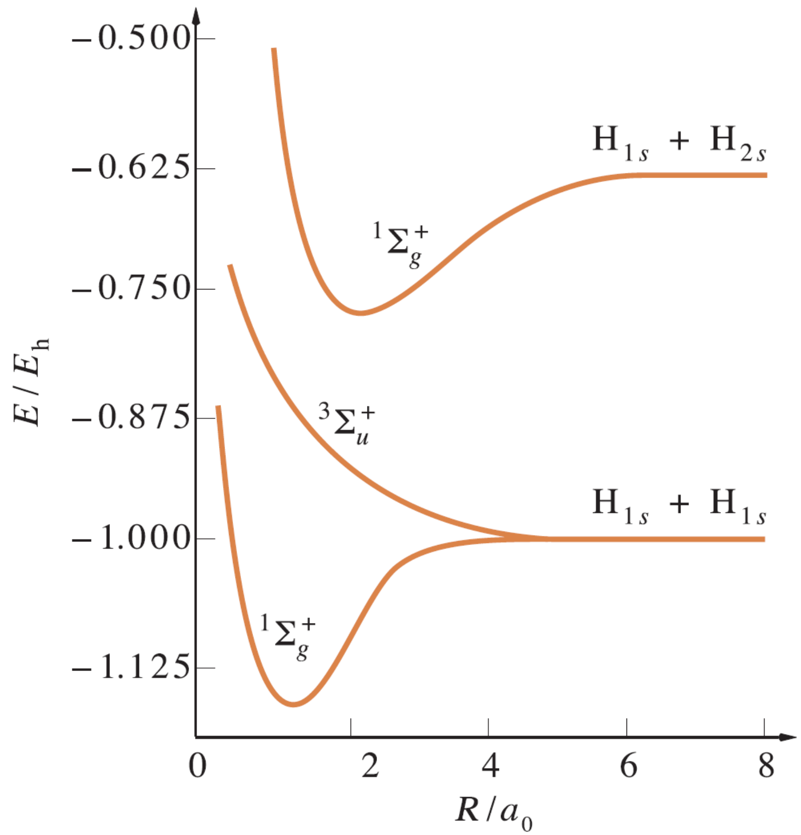


- Both orbitals are *u*, so $u \cdot u = g$
- One is (+) and the other is (-), so (+)(-) = (-)
- Hence, the ground-state term symbol is ${}^3\Sigma_g^-$

(Image: McQuarrie, ***Quantum Chemistry***, 2nd Ed., Figure 11.11)

M 11.5

Potential Energy Curves



(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Figure 11.12)

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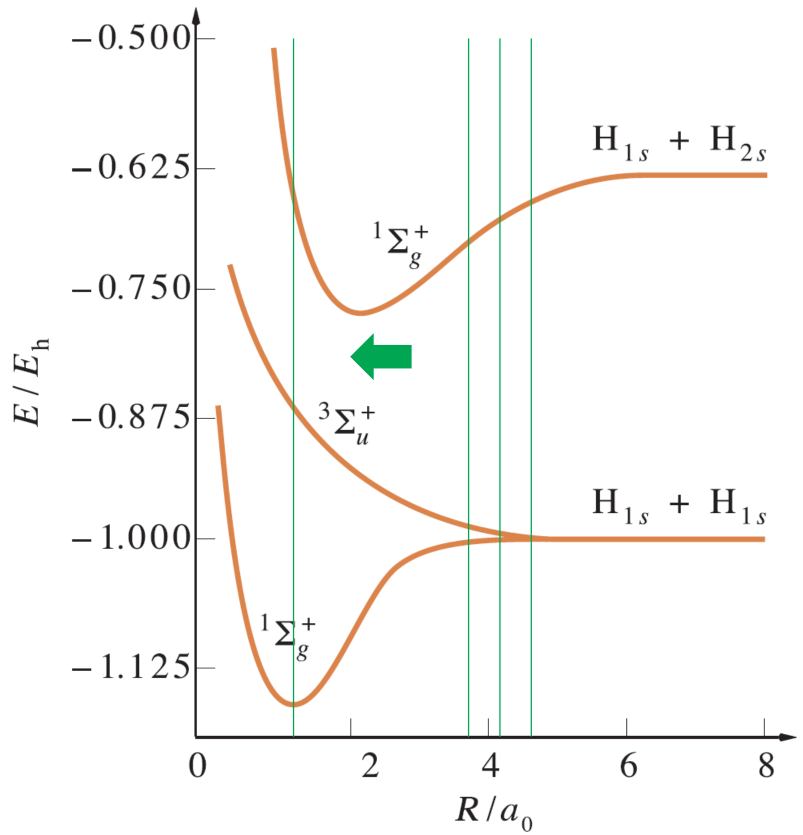
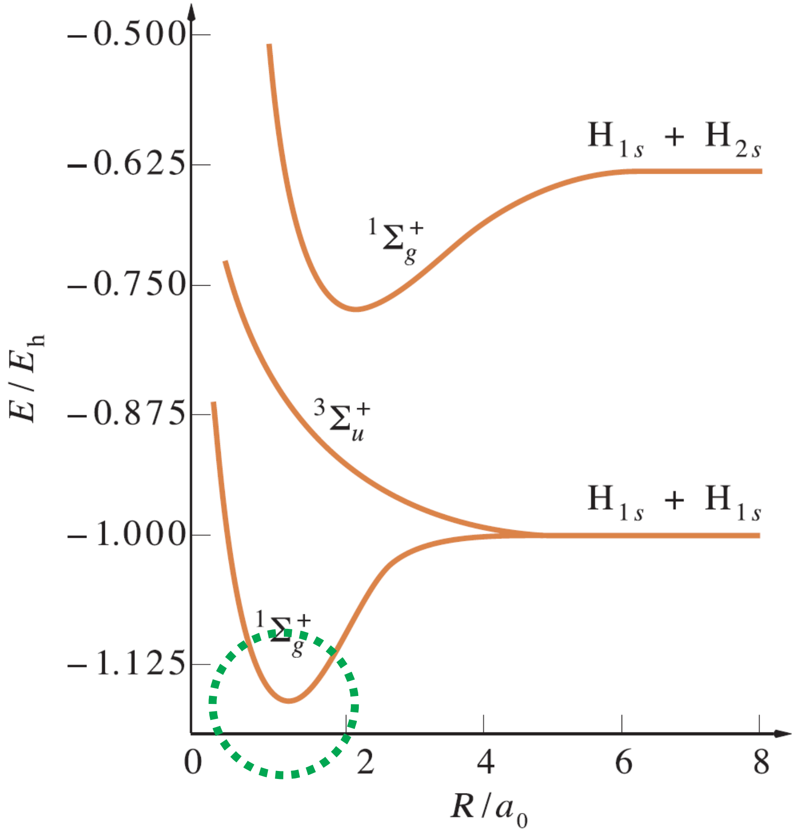
Geometry Optimization



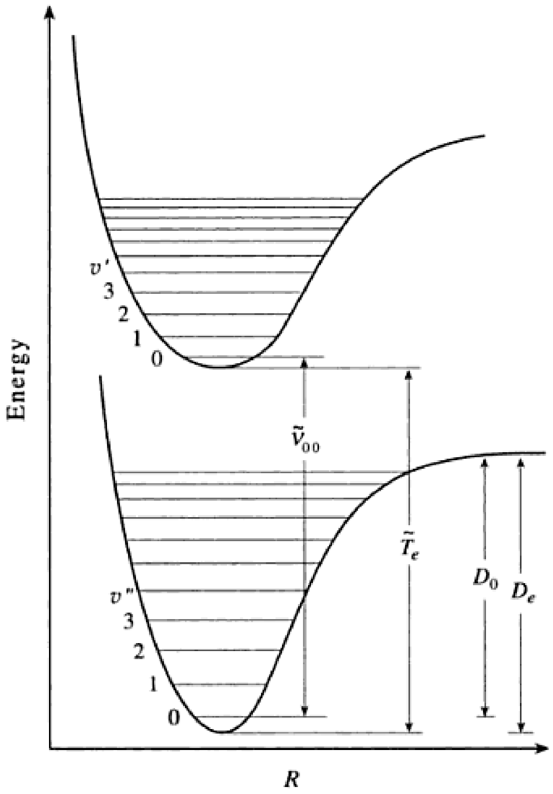
(Image: McQuarrie, ***Quantum Chemistry***, 2nd Ed., Figure 11.12)

Origin of Molecular Vibrations



(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Figure 11.12)

Vibrational Levels



(Image: https://chem.libretexts.org/@go/page/2399)

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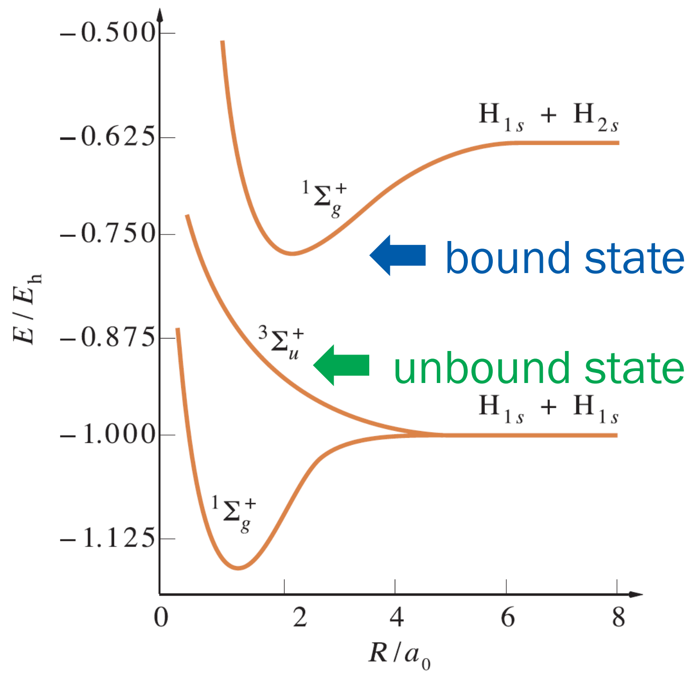
Fate of Excited System

depends on the excited state: bound or unbound?



(Image: McQuarrie, ***Quantum Chemistry***, 2nd Ed., Figure 11.12)

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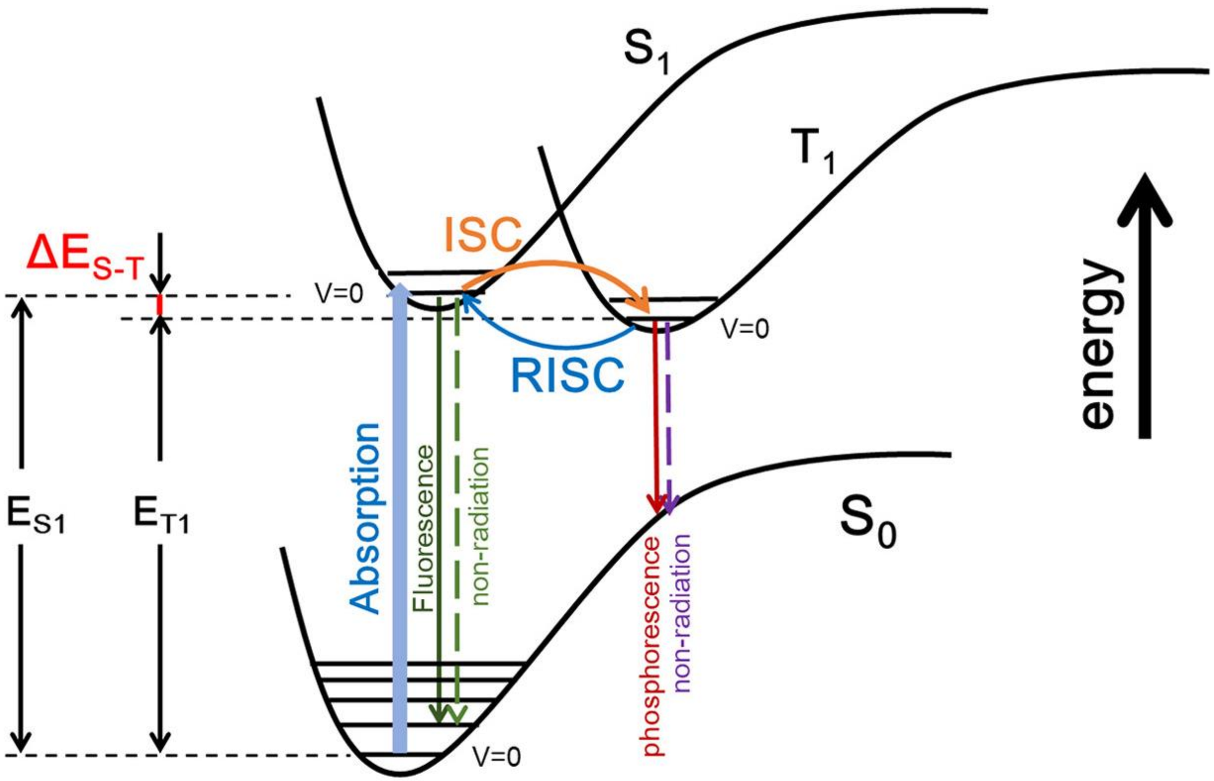
Fluorescence and Phosphorescence



(Image: Zheng *et al.*, ***J. Phys. Chem. Lett.***, 2022)

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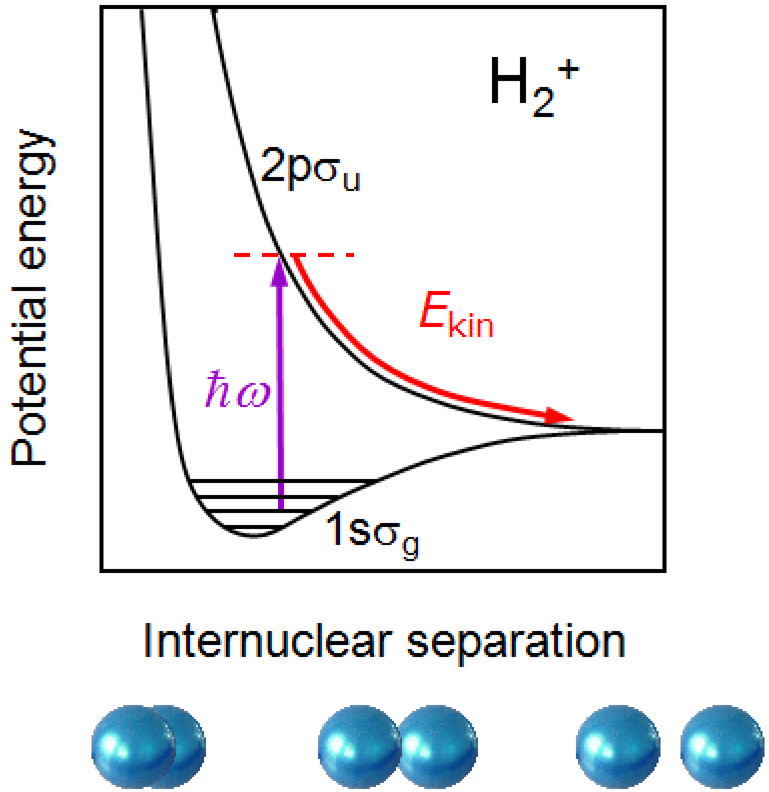
Photodissociation



(Image: https://commons.wikimedia.org/wiki/File:Hydrogen_molecular_ion_dressed_in_photons.png)

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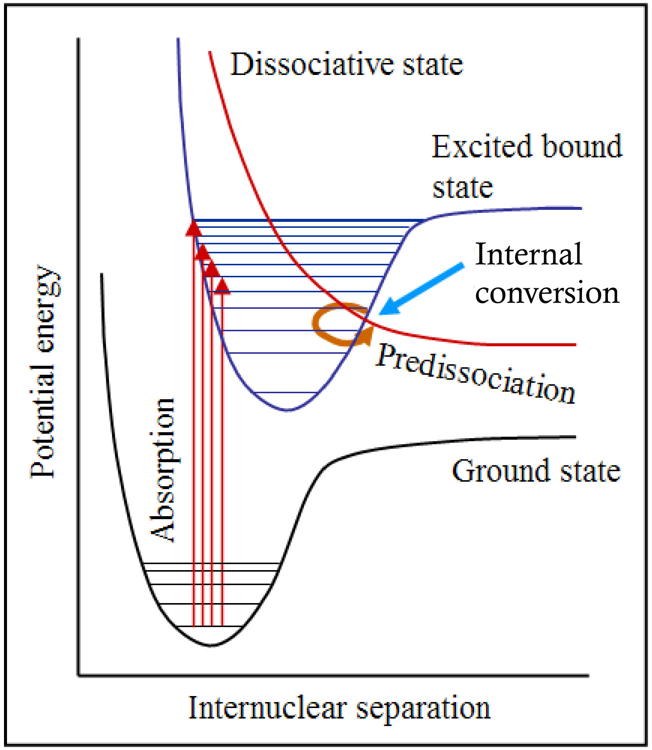
Predissociation



(Image: Iqbal, Ph.D. Dissertation, 2010)